ASSIGNMENT 4

TASK 2 — SPOKEN ARABIC DIGIT SEQUENTIAL CLASSIFICATION

Recurrent Neural Networks, LSTM, GRU and Bonus Experiments

Course: Deep Learning

Dataset: UCI Spoken Arabic Digit

Input representation: 13-dimensional MFCC sequences

Implementation: Python and PyTorch

Report prepared from the submitted notebook and its recorded outputs

1. Introduction

This task studies spoken-digit classification using sequential Mel-Frequency Cepstral Coefficients (MFCCs). Each spoken utterance is represented as a variable-length sequence of acoustic feature vectors. The notebook implements Vanilla RNN, LSTM and GRU classifiers, compares recurrent depth and hidden width, and evaluates bonus architectures using bidirectional recurrence and temporal attention pooling. The experiments also record training-loss curves, confusion matrices and gradient diagnostics.

2. Dataset and Sequential Representation

The notebook uses the UCI Spoken Arabic Digit dataset, loaded from Train_Arabic_Digit.txt and Test_Arabic_Digit.txt. The recorded data contains 6,600 training utterances and 2,200 held-out test utterances. There are ten classes (digits 0–9), with 660 training utterances and 220 test utterances per class. Each frame has 13 MFCC coefficients. Sequence lengths range from 4 to 93 frames, with a recorded mean of 39.8 frames.

An utterance is represented as $X = [x_1, x_2, \ldots, xT]$, where each $x_t \in R^{13}$ and T varies by utterance. The notebook uses padded batches with packed recurrent sequences so padded frames do not influence the recurrent state. MFCC coefficients are standardized using statistics calculated from training frames only.

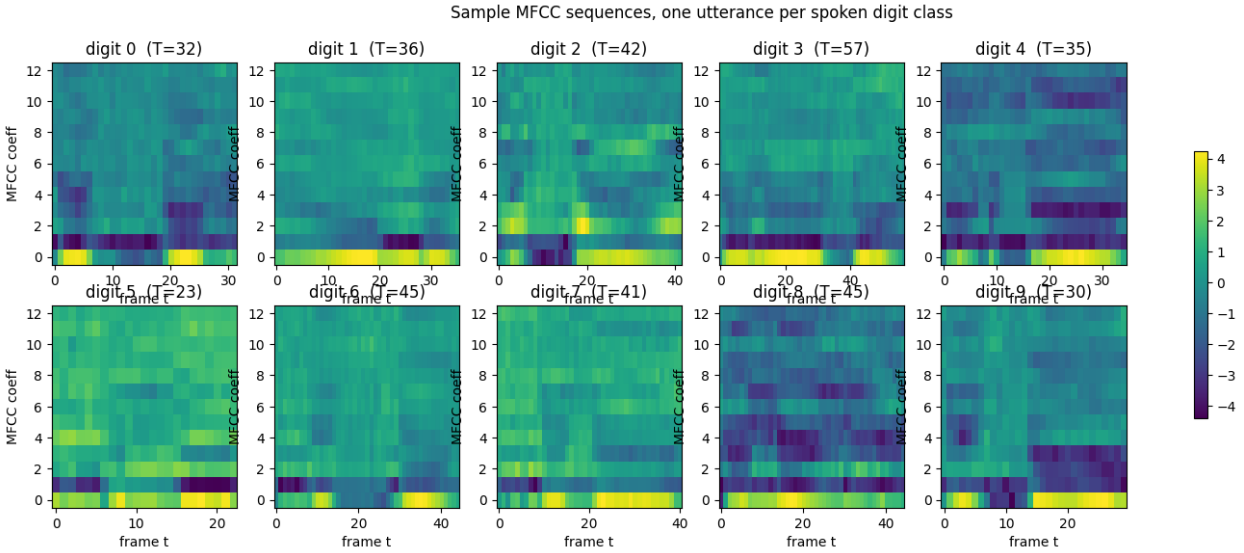


Figure 1. One representative MFCC heatmap for each of the ten spoken digit classes.

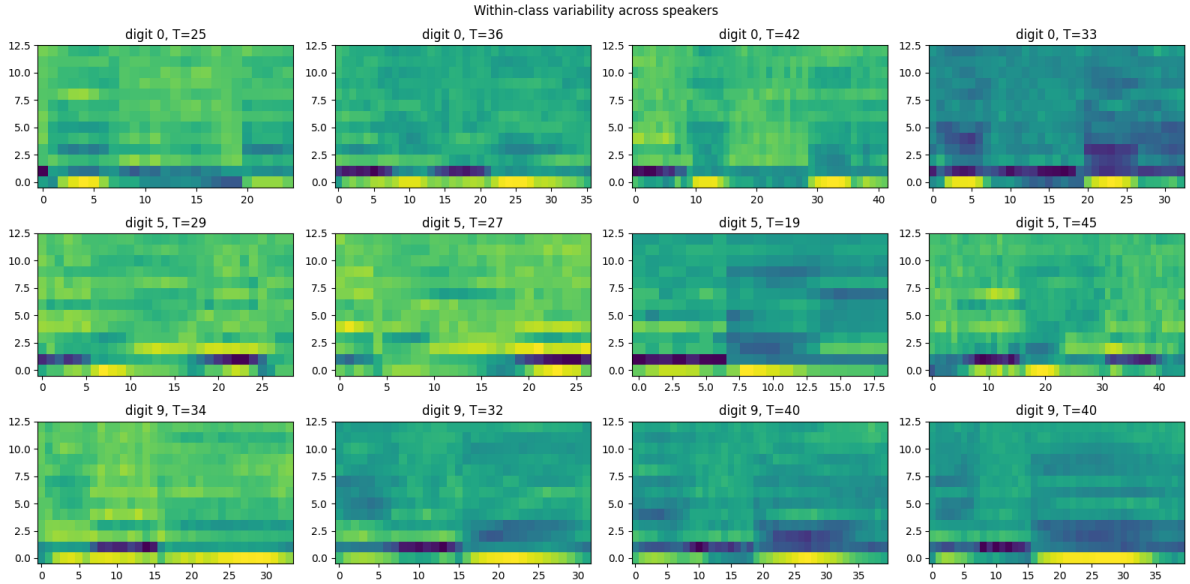


Figure 2. Examples illustrating within-class variation across utterances.

3. Data Partitioning and Preprocessing

The official training file is split into 5,940 training utterances and 660 validation utterances using a stratified 90%/10% split with random seed 42. The official test file is kept separate and contains 2,200 utterances. This preserves a held-out evaluation set and maintains equal class counts. Feature means and standard deviations are computed from training frames only, then applied to training, validation and test sequences.

Partition	Utterances	Purpose
Training	5,940	Parameter updates
Validation	660	Model selection and validation monitoring
Held-out test	2,200	Final evaluation

4. Model Architecture

The model consumes 13 features per timestep and produces ten class logits. The baseline encoder is one of three recurrent cells: Vanilla RNN, LSTM or GRU. The hyperparameter grid varies the number of recurrent layers (1 or 2) and hidden size (32, 64 or 128), producing 18 baseline configurations. For non-attention models, the last valid recurrent state is used as the sequence representation. The optional attention variant scores valid timestep outputs, applies a softmax over time, and forms a weighted sequence representation.

5. Training Procedure and Convergence

The notebook uses Adam with learning rate 0.001, batch size 64, gradient clipping at 1.0, and a fixed seed of 42. The convergence criterion is $|\bar{L}(k) - \bar{L}(k-1)| < 10^{-4}$, where $\bar{L}$ is the recorded average training loss for an epoch. The maximum of 150 epochs is a safety cap, not the intended stopping rule. Runs that reach the cap are marked as not converged by the implemented criterion.

Implementation note: the training loss is averaged over batches rather than weighted by the number of examples in each batch. The reported convergence flags therefore reflect the notebook's recorded calculation.

6. Hyperparameter Search Results

Eighteen baseline configurations were tested. Table 1 reproduces the notebook's saved results. Accuracies are percentages; parameter counts and times are as recorded in the execution.

Model	Params	Epochs	Converged	Train %	Val %	Test %	Final loss	Time (s)
RNN-L1-H32	1834	150	No	96.06	93.03	86.50	0.1238	81.2
RNN-L1-H64	5706	26	Yes	94.66	92.73	86.91	0.2334	14.1
RNN-L1-H128	19594	150	No	99.14	97.73	91.68	0.0764	81.0
RNN-L2-H32	3946	31	Yes	92.39	90.00	83.14	0.3445	24.6
RNN-L2-H64	14026	150	No	99.04	96.97	92.41	0.0245	117.5
RNN-L2-H128	52618	150	No	99.71	98.94	94.59	0.0209	118.3
LSTM-L1-H32	6346	90	Yes	100.00	99.39	95.32	0.0005	55.2
LSTM-L1-H64	20874	52	Yes	100.00	99.09	95.68	0.0004	31.1
LSTM-L1-H128	74506	31	Yes	100.00	99.24	97.50	0.0004	18.4
LSTM-L2-H32	14794	54	Yes	100.00	99.55	95.73	0.0005	45.5
LSTM-L2-H64	54154	22	Yes	100.00	99.39	95.82	0.0007	18.5
LSTM-L2-H128	206602	18	Yes	100.00	99.55	96.18	0.0005	16.4
GRU-L1-H32	4842	28	Yes	99.78	98.33	93.64	0.0209	16.0
GRU-L1-H64	15818	43	Yes	100.00	98.94	96.36	0.0007	24.1
GRU-L1-H128	56202	30	Yes	100.00	99.24	96.73	0.0006	16.9
GRU-L2-H32	11178	34	Yes	99.88	99.39	95.73	0.0017	27.5
GRU-L2-H64	40778	24	Yes	100.00	99.39	95.05	0.0008	20.0
GRU-L2-H128	155274	21	Yes	100.00	99.39	96.82	0.0005	17.7

7. Training-Loss Curves

The following plots compare average training loss across every tested configuration and overlay configurations within each recurrent cell family. Several Vanilla RNN runs reached the 150-epoch cap, whereas all listed LSTM configurations and most GRU configurations met the implemented loss-difference criterion earlier.

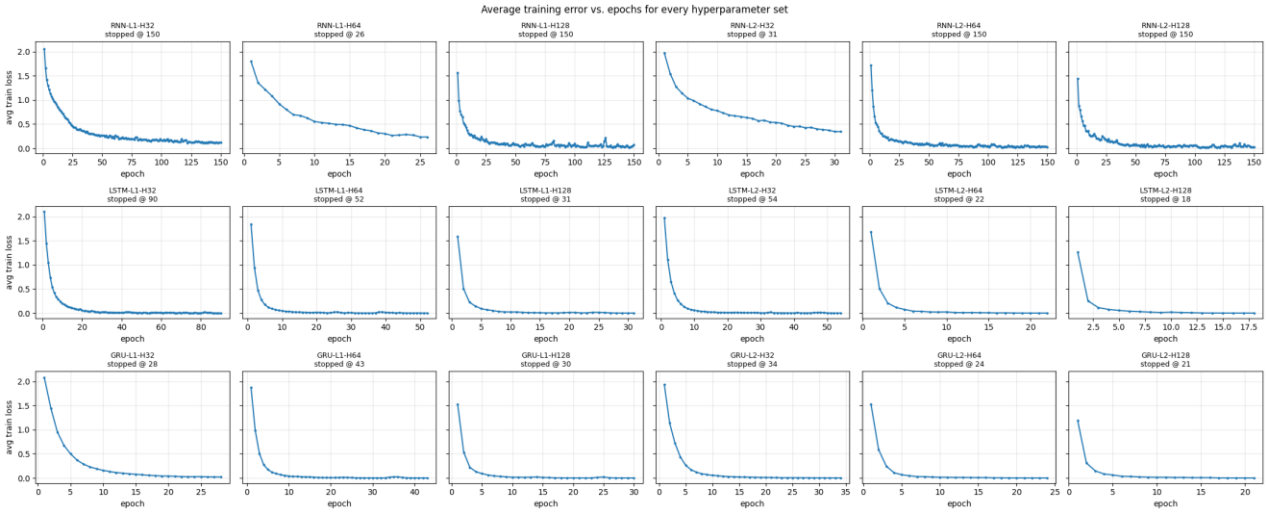


Figure 3. Average training loss versus epochs for all 18 baseline configurations.

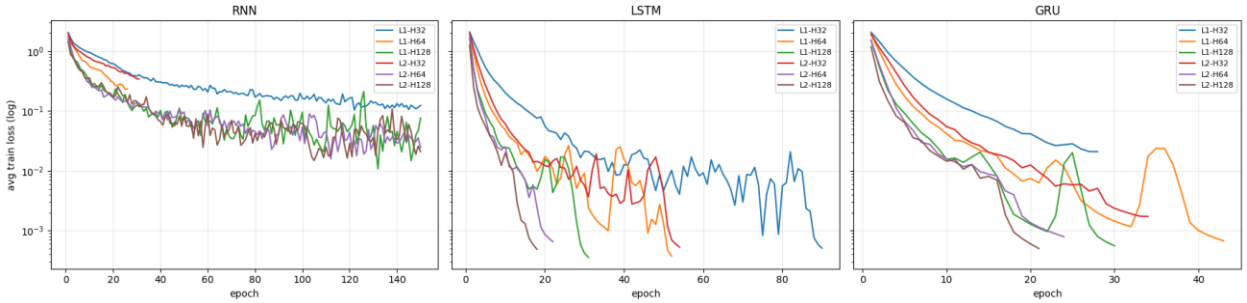


Figure 4. Training-loss curves overlaid by recurrent cell family.

8. Best Baseline Model and Test Metrics

The best baseline is selected using validation accuracy rather than test accuracy. The notebook selects LSTM-L2-H32 (two LSTM layers, 32 hidden units), with 100.00% training accuracy, 99.55% validation accuracy, and 95.73% held-out test accuracy after 54 epochs. Its classification report gives macro precision 95.92%, macro recall 95.73%, and macro F1-score 95.74%.

Metric	Recorded value
Selected baseline	LSTM-L2-H32
Epochs	54
Training accuracy	100.00%
Validation accuracy	99.55%
Test accuracy	95.73%
Macro precision	95.92%
Macro recall	95.73%
Macro F1-score	95.74%

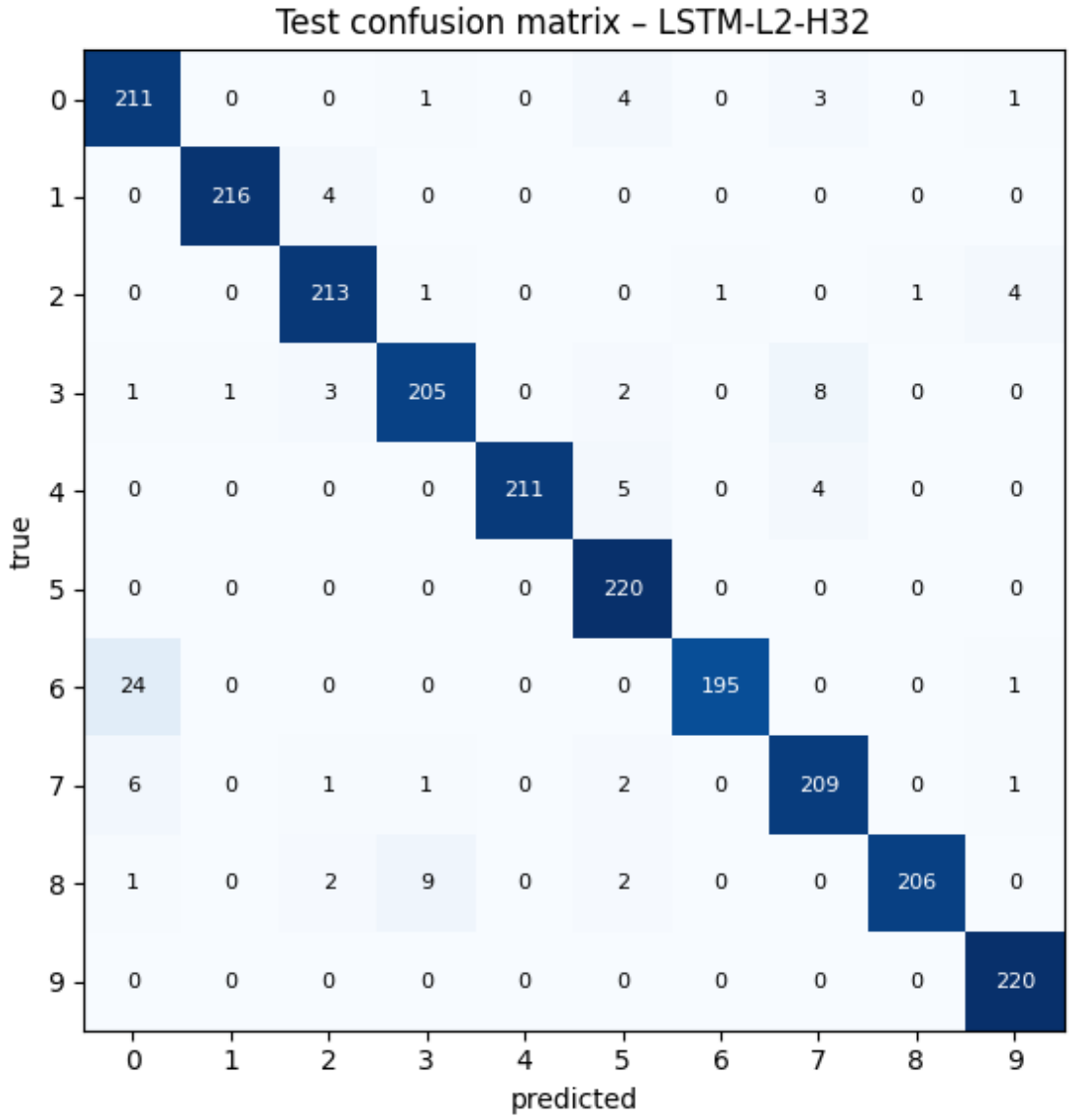


Figure 5. Test confusion matrix for the selected baseline LSTM-L2-H32.

The test set contains 2,200 utterances, with 220 examples per class. Per-class recall is highest for digits 5 and 9 (100.00%) and lower for digit 6 (88.64%) and digit 0 (95.91%). Digit 0 has the lowest recorded precision (86.83%), suggesting that predictions of digit 0 account for a notable share of errors.

9. Bonus: Bidirectional Recurrence and Temporal Attention

The bonus experiments start from the selected LSTM-L2-H32 baseline and compare a bidirectional LSTM, temporal attention pooling, and the combination of both. Table 2 gives the recorded values.

Configuration	Params	Epochs	Train %	Val %	Test %	Final loss
LSTM-L2-H32	14794	54	100.00	99.55	95.73	0.0005
LSTM-L2-H32-Bi	37770	24	100.00	99.24	96.77	0.0008
LSTM-L2-H32-Attn	15883	40	99.95	99.09	96.27	0.0025
LSTM-L2-H32-Bi-Attn	41995	16	99.90	99.55	95.64	0.0048

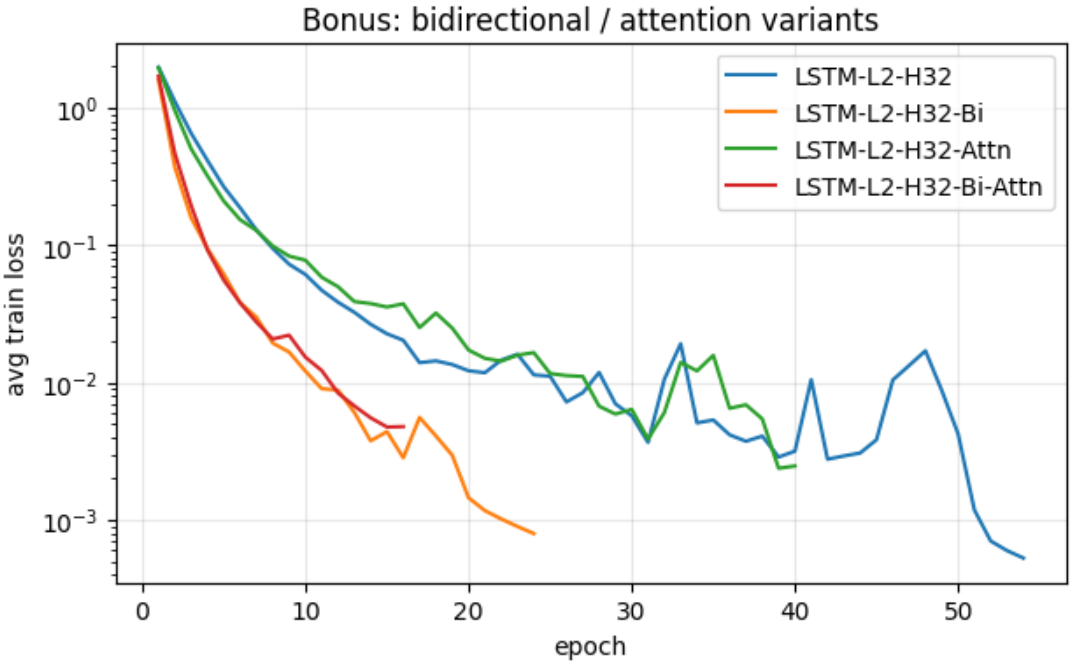


Figure 6. Training-loss comparison for the baseline and bonus variants.

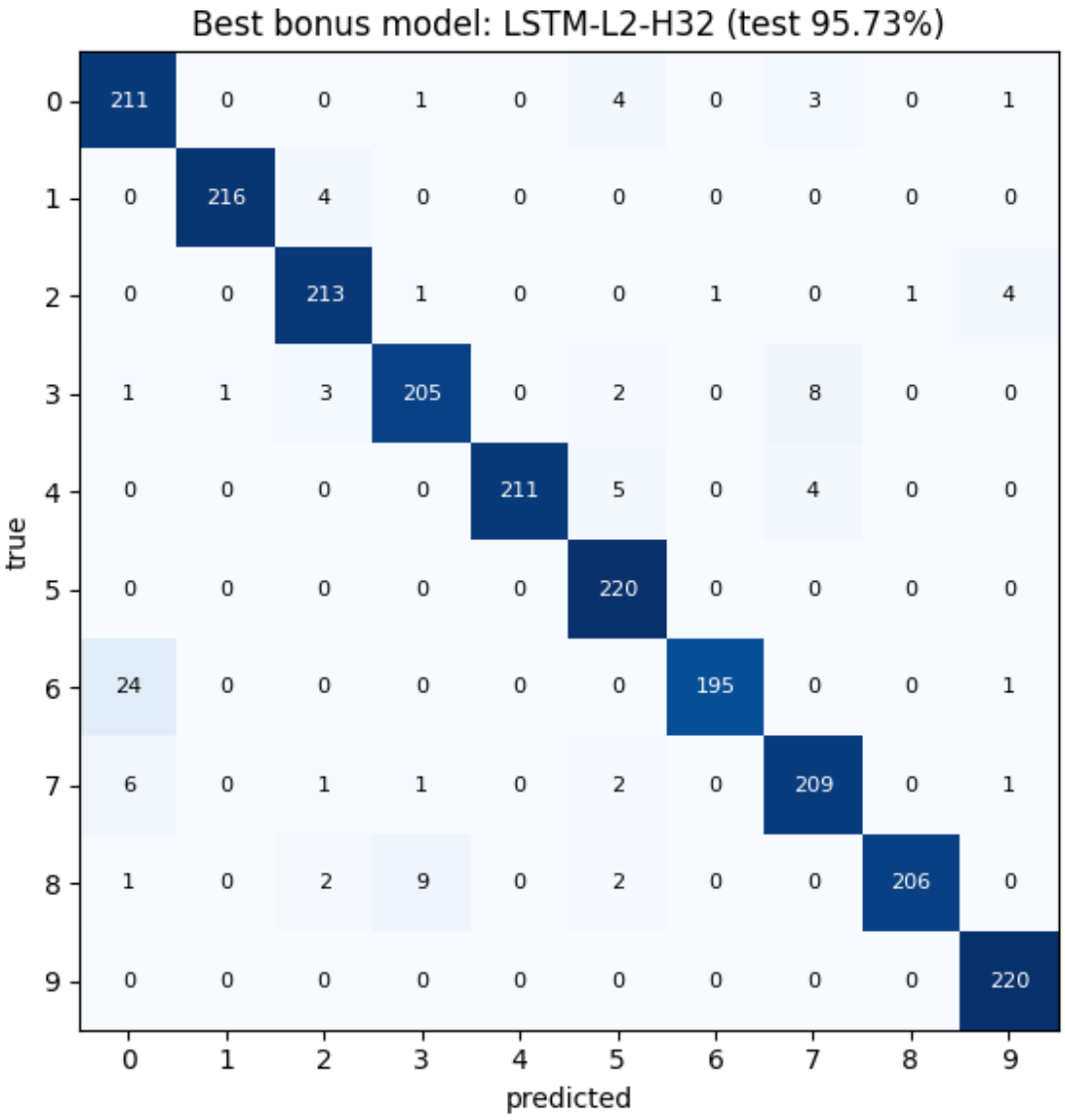


Figure 7. Confusion matrix for the best bonus model selected by validation accuracy.

The bidirectional LSTM reached 96.77% test accuracy, compared with 95.73% for the baseline. LSTM with attention reached 96.27%, while the combined bidirectional-attention model reached 95.64%. The combined model tied the baseline's validation accuracy at 99.55%, but did not improve the test score. These outcomes show that bonus mechanisms are not automatically additive. Selection remains based on validation accuracy, and the held-out test set is used to report final generalization.

10. Gradient-Flow Analysis

The notebook records mean pre-clipping gradient norms during training and also estimates the input-gradient profile as a function of distance from the final timestep. The recorded relative input-gradient norms are:

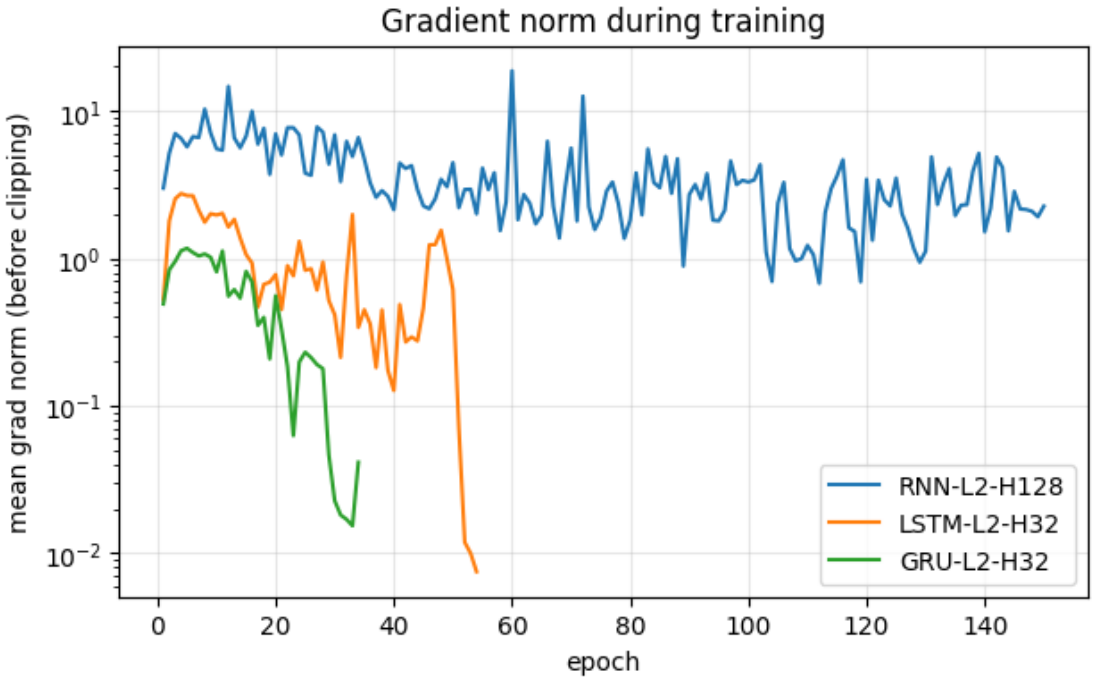


Figure 8. Mean gradient norm during training for the best configuration of each cell type.

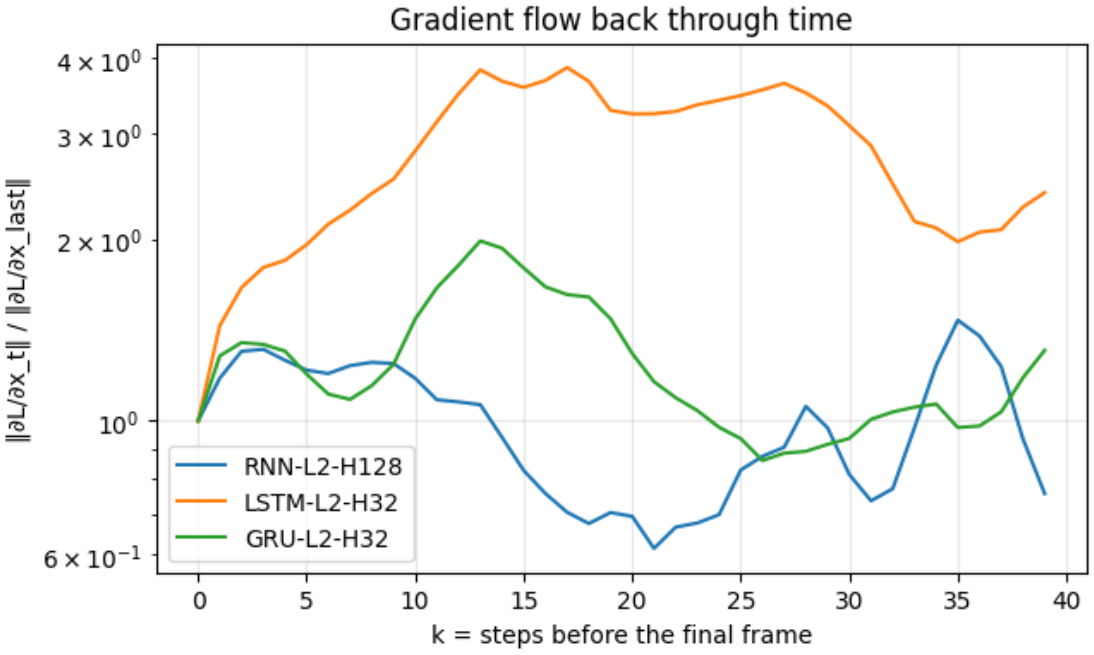


Figure 9. Input-gradient magnitude relative to the final frame, by distance through time.

Cell family	Relative gradient k=10	k=20	k=39
Vanilla RNN	1.177	0.6944	0.7578
LSTM	2.806	3.232	2.393
GRU	1.476	1.294	1.310

In this diagnostic, the LSTM and GRU input-gradient profiles retain larger relative magnitudes at earlier timesteps than the Vanilla RNN profile. This is consistent with gating mechanisms helping preserve gradient flow through time. These values are empirical for the selected models and diagnostic; they should not be interpreted as a universal guarantee that gradients never vanish or explode.

11. Comparative Discussion

Across the baseline grid, the best recorded test accuracy by family is 94.59% for Vanilla RNN, 97.50% for LSTM, and 96.82% for GRU. Mean test accuracy across each family is 89.20%, 96.04%, and 95.72%, respectively. The Vanilla RNN family also has a higher mean epoch count (109.5) and several runs that reached the 150-epoch cap, while the mean counts are 44.5 for LSTM and 30.0 for GRU. This suggests the gated models learned this task more reliably under the chosen settings.

The best baseline achieves perfect training accuracy but 95.73% test accuracy, indicating a generalization gap. The test scores across models are not monotonic in hidden width or depth: additional units or layers do not guarantee improved held-out performance. Model selection on validation accuracy is therefore important.

The notebook reports that 78% of baseline runs met the 10^{-4} criterion before the epoch cap. Loss differences between adjacent epochs are a practical stopping signal, but they do not guarantee optimal validation or test performance. The implementation also reports batch-averaged losses, so convergence should be understood according to that recorded definition.

12. Conclusion

The experiment models spoken Arabic digit recognition as classification over variable-length sequences of 13-dimensional MFCC vectors. Among the baseline families, LSTM has the highest best test accuracy in the saved results, with GRU close behind and Vanilla RNN lower. The bidirectional LSTM bonus variant records the highest test accuracy among the four reported baseline/bonus comparisons (96.77%), although the best baseline remains selected by validation accuracy in the notebook. The MFCC heatmaps, loss curves, confusion matrices, and gradient diagnostics provide complementary views of sequence structure, predictive performance, convergence, and gradient propagation.